

## Chapter 12

# Continuous Random Variables: pdf, cdf, Percentiles

Lecture 12 | Devore 9e: 4.1–4.2

### 12.1 What goes wrong with a pmf

A random variable  $X : S \rightarrow \mathbb{R}$  is **discrete** if its values form a finite or countable set, and **continuous** if they fill intervals —  $[a, b]$ ,  $[a, \infty)$ , or a union such as  $(a, b) \cup (c, \infty)$ .

**Example 12.1** (Height in the Himalayas). Pick a point at random in a mountain region and let  $X$  be its elevation, so  $X \in [0 \text{ ft}, 29\,029 \text{ ft}]$ . What is  $\mathbb{P}\{X = 10\,000 \text{ ft}\}$ ?

Exactly 10 000.000... feet, to infinite precision, is a contour line of zero area. The answer is 0. And the same argument applies to *every* value:

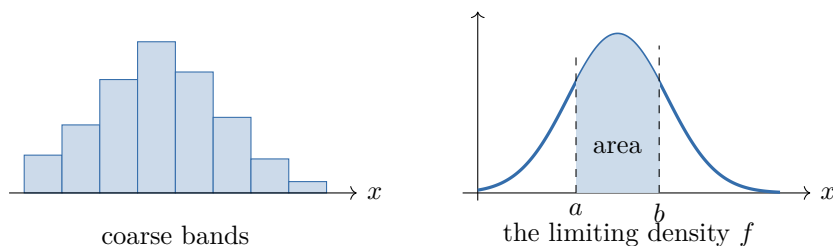
$$\mathbb{P}\{X = c\} = 0 \quad \text{for every } c.$$

So a pmf is useless here — it would be identically zero. But the variable is certainly not trivial: some elevation bands are far more likely than others. The information lives in *intervals*, not in points.

**Definition 12.2** (Continuous random variable).  $X$  is **continuous** if its possible values form intervals and  $\mathbb{P}\{X = c\} = 0$  for every  $c \in \mathbb{R}$ .

### From histogram to density

Chop the range into bands  $[0, 100]$ ,  $(100, 200]$ ,  $(200, 300]$ , ... and record the probability of each. That gives a histogram. Now refine the bands: each bar gets shorter, so instead plot *probability per unit length*, i.e. bar height divided by bar width. As the width goes to 0 the tops of the bars trace out a curve. That curve is the density.



## 12.2 The probability density function

**Definition 12.3** (pdf). The **probability density function** of a continuous random variable  $X$  is a function  $f$  such that for all  $a < b$ ,

$$\mathbb{P}\{a \leq X \leq b\} = \int_a^b f(x) dx.$$

**Theorem 12.4** (Properties).

$$f(x) \geq 0 \text{ for all } x, \quad \int_{-\infty}^{\infty} f(x) dx = 1.$$

*Conversely, any  $f$  with these two properties is the pdf of some continuous random variable.*

Two consequences worth stating separately.

**Proposition 12.5.**  $\mathbb{P}\{X = c\} = 0$  for every  $c$ , and therefore

$$\mathbb{P}\{a \leq X \leq b\} = \mathbb{P}\{a < X \leq b\} = \mathbb{P}\{a \leq X < b\} = \mathbb{P}\{a < X < b\}.$$

*Proof.*  $\mathbb{P}\{X = c\} = \lim_{h \rightarrow 0} \int_c^{c+h} f(x) dx = 0.$  □

**Remark 12.6** (The crucial warning).

$$\boxed{f(x) \neq \mathbb{P}\{X = x\}}$$

A pdf is *not* a probability. It is a probability *per unit length*, so it can exceed 1 —  $f(x) = 5$  on  $[0, 0.2]$  is a perfectly good density. What  $f$  measures is relative likelihood: for small  $\delta$ ,

$$\mathbb{P}\{c - \delta \leq X \leq c + \delta\} = \int_{c-\delta}^{c+\delta} f(x) dx \approx 2\delta f(c),$$

so a large  $f(c)$  means  $X$  is likely to land *near*  $c$ , not *at*  $c$ .

| Discrete                                  | Continuous                            |
|-------------------------------------------|---------------------------------------|
| values countable                          | values fill intervals                 |
| pmf $p(x) = \mathbb{P}\{X = x\}$          | pdf $f(x) \neq \mathbb{P}\{X = x\}$   |
| $\sum_i \mathbb{P}\{X = x_i\} = 1$        | $\int_{-\infty}^{\infty} f(x) dx = 1$ |
| $\mathbb{P}(A) = \sum_{x_i \in A} p(x_i)$ | $\mathbb{P}(A) = \int_A f(x) dx$      |

## 12.3 The cdf, again

The cdf is defined exactly as before, and it is what unifies the two cases.

**Definition 12.7** (cdf of a continuous random variable).

$$F_X(x) = \mathbb{P}\{X \leq x\} = \int_{-\infty}^x f(t) dt.$$

**Theorem 12.8.** Where  $f$  is continuous,

$$f(x) = F'_X(x),$$

by the fundamental theorem of calculus. Moreover

$$\mathbb{P}\{a < X < b\} = F_X(b) - F_X(a), \quad \mathbb{P}\{X > a\} = 1 - F_X(a).$$

**Remark 12.9.** Compare with the discrete case: there,  $F$  was a step function and the pmf was the size of the jumps. Here  $F$  is continuous and the pdf is its slope. Same object, two regimes — which is why Lecture 8 insisted that the cdf, not the pmf, is the fundamental description.

## 12.4 Percentiles

**Definition 12.10** (Percentile). For  $0 < p < 1$ , the  $(100p)$ -th percentile of a continuous random variable  $X$  is the value  $x_p$  with

$$\mathbb{P}\{X \leq x_p\} = p, \quad \text{i.e.} \quad F_X(x_p) = p.$$

So  $x_p = F_X^{-1}(p)$ : percentiles are what you get by reading the cdf backwards.

|                |                 |
|----------------|-----------------|
| median         | 50th percentile |
| first quartile | 25th percentile |
| third quartile | 75th percentile |

**Example 12.11** (Percentiles of a uniform). For  $X \sim \text{Uniform}[A, B]$  we will see below that  $F(x) = \frac{x-A}{B-A}$  on  $[A, B]$ . To find the 30th percentile, set  $F(x_{0.3}) = 0.3$  and solve:

$$\frac{x_{0.3} - A}{B - A} = 0.3 \quad \implies \quad x_{0.3} = A + 0.3(B - A).$$

So it sits 30% of the way along the interval — exactly what “uniform” should mean. The median is  $x_{0.5} = A + \frac{1}{2}(B - A) = \frac{A+B}{2}$ .

## 12.5 Two examples

**Example 12.12** (Uniform distribution on  $[A, B]$ ). “Equally likely anywhere in  $[A, B]$ ” means constant density:

$$f(x) = \begin{cases} \frac{1}{B-A}, & A \leq x \leq B, \\ 0, & \text{elsewhere,} \end{cases} \quad F(x) = \begin{cases} 0, & x < A, \\ \frac{x-A}{B-A}, & A \leq x \leq B, \\ 1, & x > B. \end{cases}$$

The constant  $\frac{1}{B-A}$  is forced by  $\int f = 1$ . Note it exceeds 1 whenever  $B - A < 1$ .

**Example 12.13** (Throwing a dart — density is not uniform even when the dart is). A dart lands uniformly on a disc of radius 1. Let  $X$  be the distance from the center. *The dart is uniform on the disc, but  $X$  is not uniform on  $[0, 1]$ .*

Go through the cdf, which is where geometric probability from Lecture 4 pays off:

$$F(x) = \mathbb{P}\{X \leq x\} = \frac{\text{area of disc of radius } x}{\text{area of disc of radius } 1} = \frac{\pi x^2}{\pi \cdot 1^2} = x^2, \quad 0 \leq x \leq 1.$$

Differentiate:

$$f(x) = F'(x) = 2x, \quad 0 \leq x \leq 1.$$

Check:  $\int_0^1 2x \, dx = 1$ . ✓ The density increases linearly, because there is more area far from the center than near it.

**Remark 12.14.** This example is the template for the change-of-variable method in Lecture 14: to find the distribution of a *function* of a random variable, compute its cdf first and differentiate at the end. Do not try to transform densities directly.

## 12.6 Expectation in the continuous case

Every sum becomes an integral and every pmf becomes a pdf:

$$\underbrace{\sum_i x_i p(x_i)}_{\text{discrete}} \rightsquigarrow \boxed{\mathbb{E}[X] = \int_{-\infty}^{\infty} x f(x) \, dx}$$

$$\mathbb{E}[h(X)] = \int_{-\infty}^{\infty} h(x) f(x) \, dx, \quad \text{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2 = \int_{-\infty}^{\infty} x^2 f(x) \, dx - \mu^2.$$

**Remark 12.15** (Physical reading). Since  $\int f = 1$ ,

$$\mathbb{E}[X] = \frac{\int_{-\infty}^{\infty} x f(x) \, dx}{\int_{-\infty}^{\infty} f(x) \, dx},$$

which is precisely the **center of mass** of a rod with density  $f$ : the balance point of the pdf. This is a genuinely useful picture — for a symmetric density the mean is the axis of symmetry, no integration required.

**Example 12.16.** For the dart,  $f(x) = 2x$  on  $[0, 1]$ :

$$\mathbb{E}[X] = \int_0^1 x \cdot 2x \, dx = \frac{2}{3}, \quad \mathbb{E}[X^2] = \int_0^1 x^2 \cdot 2x \, dx = \frac{1}{2}, \quad \text{Var}(X) = \frac{1}{2} - \frac{4}{9} = \frac{1}{18}.$$

The average distance from the center is  $2/3$ , not  $1/2$ .

**Theorem 12.17** (Tail integral). *If  $X \geq 0$  is continuous, then*

$$\mathbb{E}[X] = \int_0^\infty \mathbb{P}\{X > x\} dx = \int_0^\infty (1 - F(x)) dx.$$

*Proof.* Write  $t = \int_0^t dx$  inside the integral for  $\mathbb{E}[X]$  and exchange the order of integration, which is allowed because the integrand is non-negative:

$$\mathbb{E}[X] = \int_0^\infty t f(t) dt = \int_0^\infty \int_0^t f(t) dx dt = \int_0^\infty \int_x^\infty f(t) dt dx = \int_0^\infty \mathbb{P}\{X > x\} dx. \quad \square$$

This is the continuous twin of the tail-sum formula of Lecture 10, and it pairs well with minima, whose tail probabilities factor.

**Example 12.18** (The largest and smallest of  $n$  uniforms). Let  $U_1, \dots, U_n$  be uniform on  $[0, 1]$ , generated by independent trials. Find the expected value of their maximum and of their minimum.

For the maximum,  $\mathbb{P}\{\max \leq x\} = x^n$  on  $[0, 1]$  (Lecture 8), so the pdf is  $nx^{n-1}$  and

$$\mathbb{E}[\max] = \int_0^1 x \cdot nx^{n-1} dx = \frac{n}{n+1}.$$

For the minimum,  $\mathbb{P}\{\min > x\} = (1-x)^n$ , and the tail integral gives

$$\mathbb{E}[\min] = \int_0^1 (1-x)^n dx = \frac{1}{n+1}.$$

A picture makes both answers easy to remember:  $n$  random points cut  $[0, 1]$  into  $n+1$  gaps, and the gaps turn out to be equally long on average,  $\frac{1}{n+1}$  each. The minimum is the first gap, and the maximum is 1 minus the last one. Lecture 15 finds the full distribution of each of the ordered points.

## 12.7 A worked example start to finish

**Example 12.19.** A continuous random variable has cdf  $F(x) = cx^2$  for  $0 \leq x \leq 2$  (and the obvious 0 and 1 outside). Find  $c$ , the pdf,  $\mathbb{E}[X]$  and  $\text{Var}(X)$ .

**Step 1: find  $c$ .** A cdf must reach 1 at the top of the range, so  $F(2) = 1$ :

$$c \cdot 2^2 = 4c = 1 \quad \implies \quad c = \frac{1}{4}, \quad F(x) = \frac{x^2}{4}, \quad 0 \leq x \leq 2.$$

**Step 2: differentiate to get the pdf.**

$$f(x) = F'(x) = \frac{2x}{4} = \frac{x}{2}, \quad 0 \leq x \leq 2,$$

and  $f(x) = 0$  elsewhere. Check:  $\int_0^2 \frac{x}{2} dx = \frac{x^2}{4} \Big|_0^2 = 1. \quad \checkmark$

**Step 3: the mean.**

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} xf(x) \, dx = \underbrace{\int_{-\infty}^0 x \cdot 0 \, dx}_{=0} + \int_0^2 x \cdot \frac{x}{2} \, dx + \underbrace{\int_2^{\infty} x \cdot 0 \, dx}_{=0} = \frac{1}{2} \cdot \frac{x^3}{3} \Big|_0^2 = \frac{8}{6} = \frac{4}{3}.$$

**Step 4: the second moment.**

$$\mathbb{E}[X^2] = \int_0^2 x^2 \cdot \frac{x}{2} \, dx = \frac{1}{2} \cdot \frac{x^4}{4} \Big|_0^2 = \frac{16}{8} = 2.$$

**Step 5: the variance.**

$$\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = 2 - \left(\frac{4}{3}\right)^2 = 2 - \frac{16}{9} = \frac{2}{9}.$$

**Remark 12.20.** Step 3 is written out in full to make one point: the integral  $\int_{-\infty}^{\infty}$  always splits at the edges of the support, and outside the support  $f = 0$  so those pieces vanish. After you have done it once, just integrate over the support — but be sure you know *what* the support is. Note also the sanity check in step 2; a pdf that does not integrate to 1 means an error upstream, and catching it there costs nothing.